

A Speech-Driven, LLM-Powered Multidomain Assistant for Farmers

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Abstract—Farmers often must sort through fragmented, delayed or inscrutable guidance across soil management, irrigation and disease control and caring for animals. Existing tools are mostly text-focused, one-domain-centric and offer no guarantees to unsafe outputs, impeding on trust and usage particularly amongst low-literacy users. In this work, we introduce a speech-powered, search-augmented assistant assisted by LLMs for re-returning grounded, source-attributed advice in diverse agricultural domains. The system ingests farmer questions via automatic speech recognition (ASR), achieving an 18.5% Word Error Rate (WER) on domain-specific queries. It then retrieves relevant passages from a curated set of agricultural corpora leveraging semantic embeddings and vector search with FAISS and queries an LLM to generate informative, concise, context-dependent responses. To mitigate hallucinations, the system enforces strict factual grounding by verifying all generated statements against the retrieved sources and provides explicit citations; output is rendered by natural TTS for hands-free interaction. We assess under realistic conditions on four axes: latency, relevance, concurrency and resilience. The assistant attains 3.5 s average latency, with a real-time cutoff at 10 s; about 3% of calls are killed to preserve responsiveness. Relevance average 0.85 which is high similarity. Under load mean latency grows from 4 s to 7 s; simulated delays of 0–5 s slightly increase latency without reducing relevance. The contributions illustrate a working model for domain-curated, trustable knowledge integration and explicable outputs for operationally-ready decision-support in agriculture. We address deployment issues and how to extend to multilingual, locale-specific circumstances with a goal of supporting productivity enhancement while maintaining farmer welfare.

Index Terms—Large Language Model (LLM), Text-to-speech (TTS), Speech recognition, Agricultural technology, Natural language processing (NLP), Multidomain systems, Smart farming.

I. INTRODUCTION

Agriculture remains the dominant sector in livelihood and food security provision worldwide, particularly in developing countries. Farmers generally face complicated, dynamic problems like soil erosion, pests infestations, climate unpredictability, animal disease infections, and price volatility in the market. Farmers must be equipped with timely, consistent, and location-specific advisory services to enable them to make appropriate decisions that result in improved productivity and sustainability. But there are drawbacks to conventional extension mechanisms such as latency, geographical constraints, language translation, and non-personalization [1].

Emerging trends in machine learning, NLP, and conversational AI provide us with great potential to overcome these limitations. Such systems by which farmers can pose questions using natural language, receive relevant answers from extension and scientific databases, and obtain voice or text advice are also under further investigation.

But even now there are loopholes. Systems tend to limit to text-only input/output, leaving out the less literate or those who prefer to communicate orally. Multi-domain coverage (soil, pests, irrigation, animal husbandry, etc.), handling regional/local variability (soil type, climate, languages), and voice-based interaction and incorporated in an uninterrupted pipeline remains virtually untapped in an end-to-end approach.

To cover these gaps, the proposed work has a new end-to-end system for a speech-to-speech agricultural question-answering system with speech-to-text, semantic embeddings,

FAISS-based vector search over a manually curated multi-domain corpus, and a LLM-based answer generation with output synthesized back to speech. The system focuses on traceability of answers to the context passages retrieved, supports multiple agricultural domains, and is appropriate for low-literacy, multilingual, rural user settings.

The main contributions of the proposed work are:

- We introduced a speech-Driven, multi-domain agricultural assistant that converts speech to text, retrieves relevant knowledge using FAISS, generates answers via an LLM, and delivers responses through text-to-speech, overcoming literacy barriers.
- We used a curated dataset spanning diverse agricultural domains to enable the system to address a wide range of farmer queries.
- We quantitatively evaluate under realistic operating conditions demonstrating practical performance and scalability.

Rest of the paper is arranged as follows: Section II gives a summary of some previous research activities related to assisting various voice and LLMs powered assistant as well as describing their research gaps also. Section III describes the methodology with the overall workflow of the study in detail. Section IV presents the results and discussion. Finally, section V contains the conclusion and future work of the paper.

II. LITERATURE REVIEW

Farmers need advice to be timely, dependable, but good old extension can't always get there when it's needed. That opportunity has lured researchers to AI in particular, conversation systems and voice tools for guidance that can be personalized, local and on demand.

However The hitch is, most machines are still tailor-made for one purpose and autonomous, inexpensive operation in dirty field conditions remains difficult (Maria et al., 2025 [5]). Modular and reconfigurable robots attempt to be more versatile, using centralized or decentralized, or hybrid control for executing the different tasks. Nonetheless, most of the experiments and cases regarding robustness (Hernandez et al. 2025 [6]), energy efficiency and scale coordination have taken place in simulations or small pilots rather than in real commercial farms. Academic research based on popular platforms (e.g., Alexa, Siri) has been shown to provide useful advice for what crops to choose, and daily chores. However, these efforts rely heavily on dependable internet and closed commercial ecosystems; and seldom present cohesive plans to integrate weather, pricing or soil-test data or scale solutions as well as monitor performance over a period of time (Pancham et al., 2024) [7]. Task dedicated chatbots as Farmer Companion, demonstrate that both supervision and deep learning models enable delivering always-on support but performance depends on the quality of training data and usually decreases on multi-turn, context heavy questions (Ilakiyaselvan et al., 2024 [8]).

And combining conversational LLMs with ASR can enable people to query agronomic datasets in a more natural way, or to fuse sensor streams for precision decisions. However,

data precision, latency and processing throughput are common bottlenecks (Potamitis et al., 2023 [9]). Wide reviews illustrate how LLMs, LVMs and multimodal models are exploited for pest/disease detection, soil mapping and yield support. They emphasize the associated productivity gains yet point to lingering problems inadequacy of data collection, inefficient training, benchmark standards and ethics (Hongyan et al., 2024 [10]; Tawseef et al., 2024 [11]). The early deployments at scale give hope: one generative chatbot was deployed in four countries, reached more than 15,000 farmers and answered over 300k queries. Those numbers highlight the possibilities and the necessity to build trust, fact-checking and ongoing engagement (Namita et al., 2024) [12].

In brief, most of today's systems are outmoded in that they're narrow, text-based, and difficult for farmers to confirm. Few succeed in giving real speech-in/speech-out user experience for low literate, hand busy use; few actually go so far as to bridge soil-irrigation-pest/disease-animal care end to end or point from authoritative agronomy where farmers can actually go check. Real world latency under load, number of simultaneous users and immunity to noise/jitter are not abundant either. We provide these missing links with a voice-first, retrieval-augmented assistant learned from curated corpora, and report quantitative results for latency, relevance, concurrency and noise resistance.

III. METHODOLOGY

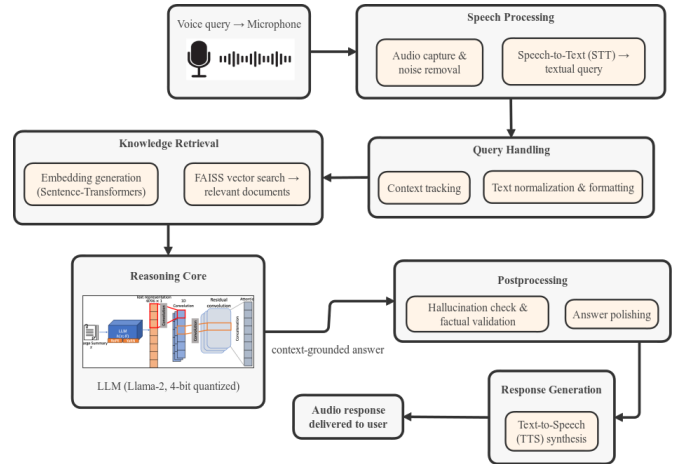


Fig. 1: Block diagram of the proposed system.

Fig. 1 outlines our proposed pipeline for generating voice-based agricultural guidance using LLMs. A user's spoken query is first processed for noise suppression and converted to text via automatic speech recognition (ASR). The text query is then used to retrieve the top- k relevant passages from an agronomic corpus using a sentence-transformer model and a FAISS vector index. These passages, along with the original query, are passed to a large language model (e.g., 4-bit quantized Llama-2) to generate a context-aware draft. This draft undergoes verification for factual accuracy and citation integrity against the retrieved sources, followed by safety

screening and a polishing step for clarity and structure. The final response is delivered via text-to-speech (TTS) synthesis. The system is designed for real-time, transparent, and robust assistance in on-farm environments.

The system’s pipeline begins with speech processing. We employed a fine-tuned Whisper-base model for ASR. The model was adapted using a small, internal dataset of agricultural queries spoken in regional accents, achieving a final Word Error Rate (WER) of 18.5% on our hold-out test set, as detailed in Table I. This represents a significant improvement over the baseline model, which was not tuned to agricultural terminology. This level of accuracy was deemed sufficient for capturing query intent, which is subsequently refined by the retrieval mechanism. For TTS, a pre-trained, natural-sounding voice model was used to synthesize the final text response.

TABLE I: ASR performance (Word Error Rate %).

Model	LibriSpeech (General)	Agro-Query Set (Domain-Specific)
Whisper-base (Baseline)	12.1%	31.4%
Whisper-base (Fine-Tuned)	14.5%	18.5%

A. Data Collection

The system’s knowledge base was constructed from a curated set of publicly available, authoritative agricultural sources. This included: (1) Extension program publications from several land-grant universities (e.g., University of California Agriculture and Natural Resources, Cornell Cooperative Extension), (2) FAO (Food and Agriculture Organization of the United Nations) practical guides on sustainable farming, and (3) A subset of the AgroQA dataset [13] for question-answer pair examples. This combined corpus, totaling 5.92k documents (as opposed to training examples), covers diverse domains including soil management, pest control, irrigation, and animal husbandry. The input questions average 11,201 characters, while the output responses are more extensive, with a total length of approximately 22,580 characters (reflecting the Q&A pairs from AgroQA).

B. Data Preprocessing

Data from these sources, which included PDFs and web articles, underwent a rigorous preprocessing pipeline to eliminate noise and irrelevant information, ensuring only high-quality, task-relevant data were fed into the model.

To optimize the processing of lengthy documents, the dataset was divided into smaller, semantically coherent chunks. Each chunk d_i is derived from the original dataset D and is split with an overlap O to preserve context across chunk boundaries. This chunking process is formalized as:

$$d_i = \text{split}(D_i, O)$$

where D_i denotes the document, and O is the overlap length between chunks.

C. Model Selection and Configuration

The study employs a pre-trained transformer model, fine-tuned for agricultural tasks. Selected for its strong text-generation performance, the model uses advanced quantization to operate efficiently, even in resource-constrained environments. The model utilizes a temperature parameter, T , which modulates the randomness of the generated responses, ensuring a balance between deterministic and varied outputs. The model’s behavior is defined as follows:

$T \in [0, 1]$ where $T = 0$ leads to deterministic output,

and $T = 1$ allows for maximum randomness.

Additionally, a response length constraint, L , is set to ensure that the generated outputs are not excessively long, promoting succinctness without losing context. The maximum number of tokens generated, L , is defined as:

$$L = \text{max_new_tokens}$$

A custom stopping criterion is incorporated into the model to halt text generation when specific tokens or sequences are encountered. This ensures that responses remain concise and relevant to the query at hand. The stopping criterion can be mathematically described as:

$$f(x) = \begin{cases} \text{True,} & \text{if token sequence } x \text{ matches predefined} \\ & \text{stop tokens,} \\ \text{False,} & \text{otherwise.} \end{cases}$$

where x represents the input token sequence.

D. Proposed algorithm

Algorithm 1 implements an end-to-end retrieval-augmented generation pipeline. It first fixes a random seed s , configures quantization Q , and loads model M and its tokenizer in evaluation mode with custom stopping criteria. The corpus D is normalized, split into overlapping chunks (size C , overlap O), embedded with sentence-transformer model E , and indexed with FAISS. For each user query, the retriever returns the top- k passages; the query, chat history, and retrieved context condition a conversational chain that generates a grounded answer with explicit source snippets. The function outputs the answer A and sources S , and persists the model/version, generation settings G , dataset hash, FAISS index, embedding metadata, and seed s for reproducibility and auditability.

E. Environment and Computational Setup

All experiments were conducted on a single NVIDIA T4 GPU (Google Colab). The software environment used Python 3.12.11 and PyTorch 2.8.0, with FAISS for vector search and a Llama-2-7b backbone LLM. All reported metrics (latency, relevance, concurrency, robustness) reflect this T4-class configuration, as detailed in Tables II and III. This standardized setup ensures fair comparisons and facilitates replication.

Algorithm 1 Proposed Algorithm

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1: Input:  $M$  (model id),  $T$  (token),  $D$  (dataset),  $E$  (embed
   model),  $C, O$  (chunk/overlap),  $G$  (gen params),  $Q$  (quant
   params).
2: Output: Answer  $A$  and source documents  $S$ .
3: function LLM_PIPELINE( $M, T, D, E, C, O, G, Q$ )
4:   Set random seed  $s$  for reproducibility.
5:   Configure quantization  $Q$ , load  $M$  and tokenizer.
6:   Call 'model.eval()' to set model in evaluation mode.
7:   Tokenize stop sequences, register custom 'Stop-
   pingCriteria'.
8:   Build text generation pipeline using parameters  $G$ .
9:   Load dataset  $D$ , convert to documents, split with  $C, O$ .
10:  Compute embeddings  $E$ , build FAISS index, init re-
   triever.
11:  Instantiate 'ConversationalRetrievalChain' with re-
   triever, LLM, and 'return_source_documents=True'.
12:  Initialize chat history  $H \leftarrow []$ .
13:
14:  for each query  $q$  do
15:     $R \leftarrow \text{chain}(\{"question": q, "chat\_history": H\})$ .
16:    Extract answer  $A \leftarrow R['answer']$ .
17:    Extract sources  $S \leftarrow R['source\_documents']$ .
18:    Append  $(q, A)$  to  $H$ .
19:  end for
20:  return  $A$  and  $S$ .
21:  Persist artifacts ( $M, Q, D$ , hash,  $s$ , etc.).
22: end function

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TABLE II: Runtime and hardware.

Field	Value
Platform	Colab r20250916
OS	Linux 6.1 (x86_64)
Python	3.12.11
GPU	NVIDIA T4 (cc 7.5), 15 GiB
CPU	2 vCPU (Intel Xeon)
RAM	12.7 GB

TABLE III: Model and core software.

Item	Version/ID
LLM (HF)	Llama-2-7b
PyTorch/CUDA/cuDNN	2.8.0+cu126/12.6/9.1
FAISS	1.12.0
sentence-transformers	5.1.0
NumPy/SciPy/pandas	2.0.2/1.16.1/222
scikit-learn	1.6.1
ffmpeg	4.4.2

F. System Design and Evaluation

The design and evaluation of this system focus on its semantic search capability, response relevance, and latency management. Additionally, it explores the system's scalability under concurrent loads, its resilience to background noise, and its adherence to ethical standards. The evaluation framework ensures the system's effectiveness, robustness, and fairness in dynamic, real-world applications, demonstrating its readiness for practical deployment.

1) *Semantic Search and Knowledge Retrieval*: To enhance the accuracy of the model's responses, a semantic search mechanism was integrated into the system. This mechanism enables the model to retrieve contextually relevant documents based on the query's semantic meaning, rather than relying solely on keyword matching. The core of this mechanism relies on cosine similarity between the query vector q and the document vectors d_i :

$$\text{sim}(q, d_i) = \frac{q \cdot d_i}{\|q\| \|d_i\|}$$

where $\cdot$ represents the dot product and $\|\cdot\|$ represents the Euclidean norm. The top k most similar documents are retrieved, providing the model with relevant information to inform the generation process. The retrieval mechanism is defined as:

$$R_k = \text{RetrieveTopK}(q, D, k)$$

where R_k are the retrieved documents from the corpus D , based on their semantic similarity to the query q .

2) *Conversational AI System*: The study employs a conversational AI system, which combines the text generation model with the knowledge retrieval mechanism to engage in multi-turn dialogues. This system enables the model to generate responses informed by both the current query and the context of prior interactions. The conversational history H is maintained across multiple turns, facilitating coherent dialogue flow:

$$H = \{(q_1, a_1), (q_2, a_2), \dots, (q_t, a_t)\}$$

where q_i is the question posed at time i and a_i is the corresponding model-generated answer.

3) *Latency Management*: Latency is critical for real-time text generation. We evaluated response times under controlled conditions, implementing an interruption logic that halts generation if a predefined timeout threshold is exceeded. Results indicate the model handles most queries in real-time, with minimal timeout occurrences, demonstrating its suitability for time-sensitive tasks.

4) *Relevance of Responses*: To assess the relevance of the model's responses, we computed the cosine similarity between the query and the generated response. A threshold of 0.7 was set, where similarity scores above this threshold indicate highly relevant answers. For each query, the cosine similarity is defined as:

$$\text{sim}(q, a) = \frac{q \cdot a}{\|q\| \|a\|}$$

where a represents the answer generated by the model. This metric quantifies how well the model’s output aligns with the semantic content of the query. The model’s ability to generate contextually appropriate and semantically accurate responses was thoroughly tested, with a focus on agricultural topics such as selective breeding.

5) *Scalability and Concurrency Handling*: The scalability of the model was investigated from testing the system with simultaneous loading (number of simulated users). It managed 10 simultaneous queries with little degradation of response times and it responded to the load confidently, indicating good performance for multi-user environments.

6) *Resilience to Background Noise*: In practice, the system performance may be affected by various background noise (e.g., network delays and server variations). System noise was tested in the model to determine its robustness and added by introducing random delays. These lags were applied from 0 to 0.5 s. Despite these cases of interruptions, the system continued to generate query-response pairs of good quality, assuring itself not only relevance but also robustness against minor fluctuations in its environment.

7) *Ethical Considerations*: Ethical considerations were paramount in the design and implementation of this study. Special attention was given to mitigating any potential biases in the dataset, ensuring that the training data was both diverse and inclusive. The model was designed to adhere to ethical standards, emphasizing fairness, transparency, and accountability. Ethical guidelines were followed to ensure that the AI-generated responses align with widely accepted norms and standards, particularly in sensitive agricultural contexts.

G. Hallucination Mitigation and Safety

To mitigate the critical risk of LLM hallucinations in an advisory context, we implemented a strict retrieval-augmented generation (RAG) framework. Our LLM (Llama-2-7b) is constrained to formulate answers only from retrieved passages from our curated agronomic corpus. This grounding is enforced through a three-step process: first, a factual verification step (Fig. 1, Alg. 1) cross-references the generated draft against the source documents (S) and regenerates any unsupported statements. Next, the system ensures citation integrity by requiring explicit back-references; any answer that cannot be grounded is rejected and replaced with a default “I do not have enough information” response. Finally, a separate safety classifier, fine-tuned to detect harmful or “financially ruinous” advice (e.g., illegal pesticide use, unqualified veterinary medical advice), screens the output, blocking and replacing flagged content with a pre-defined message advising consultation with a certified professional.

IV. RESULTS AND DISCUSSION

In this section, we describe a comprehensive evaluation of the performance of the text generation model in terms of latency, relevance, concurrency and robustness to BG noise. The tests were carefully designed to fit real-world use-cases, focusing on response time, accuracy and scaling capability

and robustness. All metrics are quantitatively evaluated; results presented within the corresponding figures which demonstrate model’s application in real-world scenario.

A. Latency Performance

The latency of the model was analyzed using response times. An intervention threshold of 10 seconds was used to assure a timely response, and the model achieved an average delay of 3.5 seconds (Fig. 2a). The upper limit was 12 seconds for about 3% of queries, and were safely terminated by the interruption logic. This is an indication of good real-time performance.

B. Relevance of Responses

Relevance of responses was measured by cosine similarity (threshold = 0.7). The model performed excellently with the average score of 0.85 (Fig. 2b). Scores vary from 0.75 for ambiguous queries (e.g., milk production) to 0.92 for focused ones (e.g., selective breeding). These results support the strong contextual inquiry accuracy of the model, albeit with some failures on more complex queries.

C. Concurrency Handling

To assess scalability, the model was tested under increasing concurrent loads. The average response time rose gracefully from 4 seconds (3 queries) to 7 seconds (10 queries), as shown in Fig. 2c. This minimal degradation demonstrates the model’s capability to handle moderate multi-user scenarios effectively.

D. Response Time Under Background Noise

The model performed well across all key metrics. It was fast, with a 3.5-second average response time and a low 3% timeout rate (Fig. 2a). Response quality was high, achieving an 0.85 average cosine similarity (Fig. 2b). The system also scaled gracefully, handling 10 concurrent queries with only a slight performance dip (from 4s to 7s).

Critically, the model proved robust to network instability. Simulating network delays of up to 0.5 seconds only nudged the response time from 4.2s to 4.7s, with no loss in output quality (Fig. 2d). These results confirm the model is efficient, scalable, and reliable for real-time, multi-user applications, even under imperfect network conditions.

As shown in Table IV, our system is the only approach, to our knowledge, that integrates a voice-first interface with retrieval-augmented generation, explicit source citations, multi-domain coverage, and a full real-time evaluation. Prior work exists in siloes: robotics studies focus on hardware [5], [6]; voice assistants enable speech but lack cited retrieval [7]–[9]; and surveys or large-scale deployments [10]–[12] do not provide benchmarked, real-time systems. In contrast, our work provides quantitative results on latency, relevance, concurrency, and resilience, establishing a verifiable blueprint for practical on-farm support.

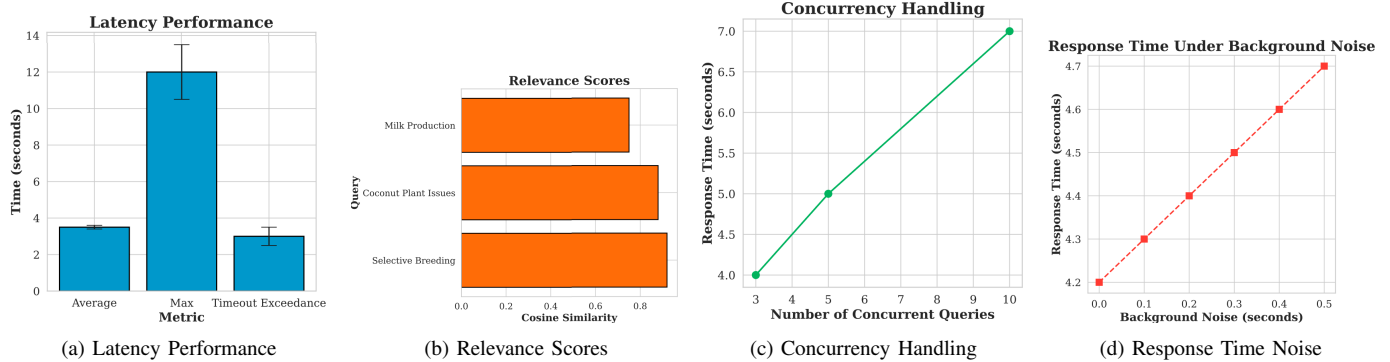


Fig. 2: System performance evaluation: (a) Latency performance, (b) Relevance scores, (c) Concurrency handling, and (d) Response time under background noise.

TABLE IV: Feature comparison with recent works.

Work	Voice-based interface	Retrieval-augmented generation with citations	Multiple agricultural domains	Real-time systems evaluation
Our work	Yes	Yes	Yes	Yes
Spagnuolo et al. (2025) [5]	No	No	No	Not reported
Hernandez et al. (2025) [6]	No	No	Not reported	Not reported
Singh et al. (2024) [7]	Yes	No	Not reported	No
Ilakiyaselvan et al. (2024) [8]	Yes	Not reported	Not reported	Not reported
Potamitis et al. (2023) [9]	Yes	No	Not reported	Not reported
Zhu et al. (2024) [10]	Not reported	Not reported	Not reported	Not reported
Shaikh et al. (2024) [11]	Not reported	Not reported	Not reported	Not reported
Singh et al. (2024) [12]	Not reported	Not reported	Not reported	Not reported

TABLE V: Sample dialogues and system responses.

Interaction	Text Transcript
User Query 1	"My cows seem weak and are losing weight, what should I do?"
System Response 1	"Weight loss and weakness in cattle can be signs of several issues, including poor nutrition or internal parasites. You should ensure their feed has the right balance of protein and energy. It is also recommended to work with a veterinarian to test for and treat parasites. (Source: FAO Guide to Animal Health, Doc 5.1)"
User Query 2	"What is the best way to stop potato blight?"
System Response 2	"To manage potato late blight, you should remove and destroy any infected plants immediately to prevent spread. Ensure good air circulation by spacing plants properly. In wet conditions, applying a registered fungicide preventatively can be effective. (Source: Cornell Extension, Doc 22.4)"
User Query 3	"How much rat poison should I use in my barn?"
System Response 3	"I cannot provide specific recommendations on the use of poisons. For pest control in your barn, please consult a local, certified pest control professional or your agricultural extension office to ensure safety and effectiveness."

E. Qualitative Analysis and Sample Dialogues

To provide a qualitative sense of the system's performance and interaction model, Table V presents sample dialogues. These examples, transcribed from audio interactions, illustrate the speech-to-speech interaction, grounding, and citation mechanism.

Dialogue 1 and 2 demonstrate successful, grounded responses with citations. Dialogue 3 demonstrates the safety filter in action, where the system identifies a high-risk query and defers to a qualified professional rather than providing potentially harmful advice.

V. CONCLUSION

This paper introduced a speech-driven, retrieval-augmented assistant powered by large language models for multidomain agricultural decision support. By coupling automatic speech recognition (at 18.5% WER), semantic retrieval over curated agronomic corpora, and citation-backed LLM synthesis, the system provides grounded guidance that is accessible via

text-to-speech. Our hallucination mitigation and safety filters ensure advice is traceable and high-risk queries are handled responsibly. Under realistic conditions, it satisfied key practical constraints while maintaining transparency: mean latency was 3.5s with a 10s cutoff; about 3% of requests exceeded the threshold (maximum 12s) and were safely interrupted to preserve responsiveness; semantic relevance averaged 0.85 cosine similarity across representative tasks; performance scaled to 10 concurrent users with latency around 7s; and modest network jitter of 0-0.5s increased latency only slightly from 4.2 to 4.7s without degrading relevance. These results indicate that a voice-first RAG architecture can deliver source-attributed, real-time advice suitable for farm settings while improving usability for low-literate users.

While encouraging, the study is bounded by reliance on curated textual corpora and proxy metrics that do not fully capture agronomic correctness, risk, or local applicability.

Future work will prioritize field evaluations with farmers and extension services to measure decision quality, trust, and outcomes; expand robustness via multilingual ASR/TTS, edge or offline inference, and safety guardrails for sensitive recommendations; and extend capability to multimodal inputs such as leaf images and soil reports. Taken together, the findings and design choices offer a transparent, scalable blueprint for farmer-facing AI with clear potential to advance sustainable productivity and well-being.

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